

Professional Summary

Electronics and Communication Engineering student with academic and project experience in FPGA-based digital design, MATLAB/Python simulation, Verilog HDL, and Xilinx Viado. Interested in electronics manufacturing, digital electronics, FPGA technology, testing, and quality. Eager to apply academic knowledge in an industrial environment while developing practical engineering skills.

Top Skills

- FPGA & Digital Design – Verilog HDL, behavioral simulation, synthesis, implementation, timing analysis and fixed-point arithmetic using Xilinx Vivado/XSim.
- Embedded & Electronics – Digital electronics, microcontroller fundamentals, STM32 basics, GPIO, ADC, PWM, timers and interrupts.
- Programming & Tools – C, C++, Python, MATLAB, Git, GitHub and VS Code.

Project Experience

FPGA Implementation of I-CDPK Codec

Final Year Project | 2026 – Present

- Developing an FPGA implementation of **Interleaved Chaotic Differential Peaks Keying (I-CDPK)** modulation and demodulation for baseband digital communication.
- Modelled **Lorenz chaotic X, Y and Z states**, peak detection, differential peak processing, interleaving, keying, and demodulation using MATLAB/Python.
- Implemented and verified FPGA-oriented modules using **Verilog HDL, Xilinx Vivado and XSim**, including behavioral simulation, synthesis, implementation, and timing analysis.
- Performed fixed-point arithmetic analysis and communication-system evaluation including **AWGN channel modelling and BER-versus-SNR analysis**.
- Used **Git and GitHub** for collaborative development, version control, documentation, and project management.

8-Bit Breadboard Computer

Digital Electronics & Computer Architecture Project | Ongoing

- Contributing to the design and construction of a **microcoded 8-bit computer** using 74LS-series TTL logic ICs.
- Implemented hardware modules including the **ALU, registers, RAM, EEPROM-based control unit, clock circuit, and shared data bus**.
- Worked with **EEPROM microprogramming and control-signal generation** for machine-level instruction execution.
- Performed hardware testing and troubleshooting involving **memory interfacing, timing, clock synchronization, and bus communication**.
- Applied practical concepts from **digital electronics, combinational and sequential logic, computer architecture, and memory organization**.

Education

Priyadarshini College of Engineering, Hingna, Nagpur

2023–2027

Bachelor of Technology – Electronics and Communication Engineering | CGPA: 7.45/10

Nagpur, Maharashtra

Class X: 79.40% **Class XII:** 53.83%

Certifications & Additional Information

- Motor Control Workshop using dsPIC® Digital Signal Controllers (DSC) Dual Core Devices [Issued by Microchip Technology/ Date issued: August 24th,2025]
- Tata – Cybersecurity Analyst job simulation [forage/ issued July 8th,2025]
- Cyber Security Course – [Introduction to cybersecurity Issued by Cisco/ Date issued: June 14th, 2024].
- Languages: English, Hindi, Marathi.

Achievements & Activities

- **1st Place – Chess Team, UDAAN 25.0 Inter-Collegiate Sports Meet (2025)**
Represented Priyadarshini College of Engineering, Nagpur, as part of the team that secured **1st place** in the chess competition held during UDAAN 25.0.

- **1st Place – Chess Team, Ashwamedh (2026)**

Represented the **ECE Department team** in the college-level chess tournament and secured **1st place**.